

ROC Curve & ROC-AUC in Machine Learning

1 Why ROC Curve?

- Many models output **probabilities**, not just class labels.
 - We need a way to see how well a model separates **positive vs. negative** classes across **all thresholds**.
 - ROC (Receiver Operating Characteristic) curve helps evaluate this trade-off.
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2 Confusion Matrix Refresher

	Predicted +	Predicted -
Actual +	TP (True Positives)	FN (False Negatives)
Actual -	FP (False Positives)	TN (True Negatives)

From it we define:

- **TPR / Sensitivity / Recall** = $TP / (TP + FN)$ → “How many actual positives we caught.”
 - **FPR** = $FP / (FP + TN)$ → “How many negatives we wrongly called positive.”
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3 Plotting ROC

- Vary the probability threshold from $0 \rightarrow 1$.
 - For each threshold, compute TPR & FPR.
 - Plot: **x-axis = FPR**, **y-axis = TPR**.
 - Start at (0,0), end at (1,1).
 - A perfect model hugs the **top-left** corner (high TPR, low FPR).
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4 Understanding Different Cases

Case	Shape of ROC
Perfect classifier	Passes through (0,1).
Good classifier	Curve above the diagonal.
Random guess	Diagonal line ($AUC \approx 0.5$).
Worse than random	Below diagonal (can flip predictions).

5 AUC (Area Under Curve)

- **AUC = Probability that a randomly chosen positive has higher score than a randomly chosen negative.**
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- Ranges 0 → 1:
 1. 1.0 = perfect.
 2. 0.5 = no discrimination.
 3. <0.5 = worse than random.
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6 Advantages

- Threshold-independent (checks all possible cut-offs).
 - Useful with **imbalanced data** (but complement with PR curve if positives are very rare).
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7 Practical Steps

1. Fit model → get predicted **probabilities**.
 2. Use `roc_curve(y_true, y_score)` in sklearn → gives FPR, TPR, thresholds.
 3. Plot FPR vs TPR.
 4. Compute `roc_auc_score(y_true, y_score)` for AUC.
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8 Key Insights

- ROC focuses on **ranking** predictions, not absolute values.
 - For highly imbalanced data, AUC-ROC may look optimistic → also check Precision-Recall curve.
 - The **steeper** the curve near (0,0), the better.
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Quick Recap

ROC shows how TPR vs FPR change as threshold moves.

AUC summarizes this: **higher AUC → better ability to rank positives over negatives.**
